

Step 1:

Kali address: 192.168.56.102

```
(kali㉿Fati)-[~/lab1-repo]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 453sec preferred_lft 453sec
    inet6 fe80::b5a4:db6f:cb15:76b6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ef:37:f8 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute eth1
        valid_lft 84753sec preferred_lft 84753sec
    inet6 fd17:625c:f037:3:9a97:fb26:20b6:5af3/64 scope global dynamic noprefixroute
        valid_lft 86199sec preferred_lft 14199sec
    inet6 fe80::1f26:46f2:8c17:78cf/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Step 2:

Target IP address: 192.168.56.103

Step 3: Screenshot confirming connectivity

```
(kali㉿Fati)-[~/lab1-repo]
$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=5.34 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.31 ms
^[[B^[[B^[[B^[[B^[[B^[[B^[[B64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.34 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=1.77 ms

— 192.168.56.103 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3048ms
rtt min/avg/max/mdev = 1.310/2.440/5.344/1.686 ms
```

Step 4: Identifying metasploitable 2

```

(kali@Fati)-[~/lab1-repo]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 05:49 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00075s latency).
MAC Address: 0A:00:27:00:00:13 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00028s latency).
MAC Address: 08:00:27:A5:99:46 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.103
Host is up (0.0018s latency).
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 6.12 seconds

```

Host discovery found four live hosts: 192.168.56.1, .100, .102, and .103. The target was confirmed as **192.168.56.103**. This was verified directly by running `ifconfig` inside the Metasploitable 2 VM, which returned this IP — matching the scan result showing an "Oracle VirtualBox virtual NIC" MAC vendor tag.

Step 5: Every Open Port

```

(kali@Fati)-[~/lab1-repo]
$ nmap 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:04 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0058s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.60 seconds

```

Step 6: Identifying Software behind Open Ports

```
(kali@Fati)-[~/lab1-repo]
$ nmap -sV 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:05 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0062s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.10 seconds
```

Step 7: Deeper Version Probing

Comparing intensity 9 against the default scan: results were the same for almost every port. The only difference was port 513, which now showed as "OpenBSD or Solaris rlogind" instead of no version at all.

This shows the default setting already worked well for most services here — going to maximum intensity only added one small detail, but took longer to run.

```

(kali@Fati)-[~/lab1-repo]
$ nmap -sV --version-intensity 9 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:09 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0046s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.94 seconds

```

Step 8: OS Detection

```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -O 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:12 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0046s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.51 seconds
```

Nmap identified the target as Linux, kernel version range 2.6.9–2.6.33 (device type: general purpose). No explicit confidence percentage was shown in this output — Nmap presented this as a direct "OS details" result rather than a list of uncertain guesses, suggesting reasonably high confidence.

This result is supported by other evidence gathered earlier: the direct login banner on Metasploitable itself showed kernel version 2.6.24-16-server, which falls within Nmap's estimated range, confirming the detection was accurate.

Step 9: Aggressive Scan

```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -A 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:16 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.56.102
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ ssl-date: 2026-09-05T17:16:34+00:00; -17h00m45s from scanner time.
|_ sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such t
hing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
| dns-nsid:
|_ bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
```

```
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version  port/proto  service
|   100000   2             111/tcp    rpcbind
|   100000   2             111/udp    rpcbind
|   100003   2,3,4         2049/tcp   nfs
|   100003   2,3,4         2049/udp   nfs
|   100005   1,2,3         52864/udp  mountd
|   100005   1,2,3         53203/tcp  mountd
|   100021   1,3,4         40784/tcp  nlockmgr
|   100021   1,3,4         46334/udp  nlockmgr
|   100024   1             36947/udp  status
|   100024   1             58448/tcp  status
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open  exec        netkit-rsh rexecd
513/tcp open  login       OpenBSD or Solaris rlogind
514/tcp open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 38
|   Capabilities flags: 43564
|   Some Capabilities: ConnectWithDatabase, SupportsCompression, Support41Auth, SupportsTransactions, LongColumnFlag, Sw
itchToSSLAfterHandshake, Speaks41ProtocolNew
|   Status: Autocommit
|_  Salt: ;+e:2*|SLCI%{'PMqp_m
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such t
hing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-09-05T17:16:33+00:00; -17h00m46s from scanner time.
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_  VNC Authentication (2)
6000/tcp open  X11        (access denied)
6667/tcp open  irc        UnrealIRCd
8009/tcp open  ajp13      Apache Jserv (Protocol v1.3)
```

```

8080/tcp open  httpd          Apache/2.2.8 (Ubuntu/11.04)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  httpd          Apache Tomcat/Coyote JSP engine 1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kern
el

Host script results:
|_clock-skew: mean: -16h00m43s, deviation: 2h00m03s, median: -17h00m45s
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_ System time: 2026-09-05T13:16:23-04:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS
1 2.04 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.85 seconds

```

The aggressive scan uncovered things a plain port scan couldn't show. Anonymous FTP login turned out to be allowed on port 21. SMB signing was found disabled on ports 139/445 — Nmap itself called this "dangerous, but default." And an SSL certificate on ports 25/5432 had expired back in 2010, with obviously fake organisation details — a clear sign this is a training system, not a real server.

Step 10:


```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -p- 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:42 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00079s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
40784/tcp open  unknown
43566/tcp open  unknown
53203/tcp open  unknown
58448/tcp open  unknown
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 25.99 seconds
```

The full port scan (all 65,535 ports) found 8 additional ports that the default scan in Step 5 missed:

- 3632/tcp — distccd
- 6697/tcp — ircs-u
- 8787/tcp — msgsrvr
- 40784/tcp — unknown
- 43566/tcp — unknown
- 53203/tcp — unknown
- 58448/tcp — unknown

Step 11:

```
(kali@Fati)-[~/Lab1-repo]
$ sudo nmap -p- -sV 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:46 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00080s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
40784/tcp open  nlockmgr     1-4 (RPC #100021)
43566/tcp open  java-rmi     GNU Classpath grmiregistry
53203/tcp open  mountd       1-3 (RPC #100005)
58448/tcp open  status       1 (RPC #100024)
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 151.85 seconds
```

Step 12: Timing Template

```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -p- -T4 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:53 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00077s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
40784/tcp open  unknown
43566/tcp open  unknown
53203/tcp open  unknown
58448/tcp open  unknown
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 28.07 seconds
```

Comparing this to Step 10, the list of open ports is exactly the same — nothing new showed up.

The scan time was also basically the same (28 seconds vs 26 seconds before).

Step 13:

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 21,22,23,25,80,139,445 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 06:57 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0081s latency).
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.36 seconds
```

Focused scanning (just picking a few ports) is useful when you already know exactly what you're looking for, instead of checking everything.

Step 14:

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 1-1024 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:00 -0400
Nmap scan report for 192.168.56.103
Host is up (0.024s latency).
Not shown: 1012 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.71 seconds
```

Step 13 only checked 7 specific ports picked, so it only showed those 7. Step 14 checks a whole range (1 to 1024), so it picked up a few extra ones that weren't in my Step 13 list, like port 53 (DNS), 111 (rpcbind), and 512/513/514. So a range scan finds more than a hand-picked list

Step 15: UDP Scan

```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -sU 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:02 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0018s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE SERVICE
53/udp    open  domain
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
111/udp   open  rpcbind
137/udp   open  netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open  nfs
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1033.71 seconds
```

Some ports showed a clean open result (53, 111, 137, 2049) — meaning the service actually replied back. Others showed open|filtered (68, 69, 138) — this means Nmap sent a request but got nothing back, so it can't tell if the port is actually open with a quiet service, or just

blocked/closed. This is the main downside of UDP scanning, no handshake means no guaranteed answer.

Step 16: Top UDP Ports + Versions

```
(kali@Fati)-[~/lab1-repo]
$ sudo nmap -sU --top-ports 20 -sV 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:29 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0018s latency).

PORT      STATE      SERVICE      VERSION
53/udp    open       domain       ISC BIND 9.4.2
67/udp    closed     dhcpd        dhcpd
68/udp    open|filtered dhcpd        dhcpd
69/udp    open|filtered tftp        tftp
123/udp   open|filtered ntp         ntp
135/udp   closed     msrpc        Microsoft Windows netbios-ns (workgroup: WORKGROUP)
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (workgroup: WORKGROUP)
138/udp   open|filtered netbios-dgm   netbios-dgm
139/udp   open|filtered netbios-ssn   netbios-ssn
161/udp   closed     snmp         snmp
162/udp   closed     snmptrap     snmptrap
445/udp   closed     microsoft-ds microsoft-ds
500/udp   open|filtered isakmp       isakmp
514/udp   closed     syslog       syslog
520/udp   open|filtered route        route
631/udp   open|filtered ipp          ipp
1434/udp  closed     ms-sql-m     ms-sql-m
1900/udp  closed     upnp         upnp
4500/udp  open|filtered nat-t-ike    nat-t-ike
49152/udp closed     unknown      unknown
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 113.52 seconds
```

Three different states came back in this scan. "Open" means the service actually replied, so we know for sure it's there — this happened on port 53 (domain) and port 137 (netbios-ns).

"Closed" means Nmap got a clear rejection back, confirming the port really isn't in use — this covered ports like 67, 135, 161, 162, 445, 514, 1434, 1900, and 49152.

"open|filtered", showed up on ports 68, 69, 123, 138, 139, 500, 520, 631, and 4500 — here Nmap got no reply at all, so it can't tell if the port is genuinely open but quiet, or blocked.

Overall, most ports gave a clear closed result, a couple were confirmed open, and the rest stayed uncertain, which fits with what's expected from UDP scanning.

Step 17: Default Scripts

```
(kali@Fati)-[~/Lab1-repo]
$ nmap -sC -sV 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:34 -0400
Nmap scan report for 192.168.56.103
Host is up (0.011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|  STAT:
|  FTP server status:
|    Connected to 192.168.56.102
|    Logged in as ftp
|    TYPE: ASCII
|    No session bandwidth limit
|    Session timeout in seconds is 300
|    Control connection is plain text
|    Data connections will be plain text
|    vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-09-05T18:34:32+00:00; -17h00m45s from scanner time.
|_ssl-v2:
|  SSLv2 supported
|  ciphers:
|    SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|    SSL2_DES_192_EDE3_CBC_WITH_MD5
|    SSL2_RC2_128_CBC_WITH_MD5
|    SSL2_RC4_128_WITH_MD5
|    SSL2_RC4_128_EXPORT40_WITH_MD5
|    SSL2_DES_64_CBC_WITH_MD5
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such th
ing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMI
ME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
```

```

111/tcp open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100003  2,3,4      2049/tcp   nfs
|_  100003  2,3,4      2049/udp   nfs
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open  exec        netkit-rsh rshd
513/tcp open  login       OpenBSD or Solaris rlogind
514/tcp open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 51
|   Capabilities flags: 43564
|   Some Capabilities: ConnectWithDatabase, Speaks41ProtocolNew, LongColumnFlag, SupportsTransactions, Support41Auth, SwitchToSSLAfterHandshake, SupportsCompression
|   Status: Autocommit
|_  Salt: 2HUPTwvrV7M^p<u\J?<N
5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-09-05T18:34:32+00:00; -17h00m46s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_  VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

```

Host script results:
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: -16h00m45s, deviation: 2h00m00s, median: -17h00m45s
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-09-05T14:34:23-04:00

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.24 seconds

```

the ftp-anon script confirmed that anonymous login is allowed on the FTP server (port 21) — meaning anyone could log in without a real username or password. Second, smb-security-mode showed that SMB message signing is disabled on ports 139/445, which Nmap itself labels as

"dangerous, but default." Both of these are things a plain port scan would never show — you only find them by actually running scripts that test the service directly.

Step 18: HTTP Title

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 80 --script http-title 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:38 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0071s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_http-title: Metasploitable2 - Linux
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.53 seconds
```

The http-title script returned: "Metasploitable2 - Linux"

Step 19: HTTP Headers

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 80 --script http-headers 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:40 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0047s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-headers:
|   Date: Sat, 05 Sep 2026 18:39:54 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 3.24 seconds
```

The Server header shows: Apache/2.2.8 (Ubuntu) DAV/2

Step 20: SMB Protocols


```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 139,445 --script smb-protocols 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:42 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0015s latency).

PORT      STATE SERVICE
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Host script results:
| smb-protocols:
|   dialects:
|_   NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 2.59 seconds
```

SMB protocol is NT LM 0.12 (SMBv1)

Step 21: SSH Host Keys

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 22 --script ssh-hostkey 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:47 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00099s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.51 seconds
```

The two SSH host key algorithms found were DSA (1024-bit) and RSA (2048-bit).

Step 22:

```
(kali@Fati)-[~/lab1-repo]
$ nmap -p 21 -sV --script banner 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-06 07:50 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ banner: 220 (vsFTPD 2.3.4)
MAC Address: 08:00:27:79:A3:4F (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.24 seconds
```

The VERSION field from -sV shows "vsftpd 2.3.4", and the banner script shows "220 (vsFTPD 2.3.4)" — the actual raw text the server sends when you connect. Both match up and confirm the

same version, just from two different methods: one is Nmap's interpreted result, and the other is the literal text straight from the service itself.

Step 23: Request Headers with Curl

```
(kali@Fati)-[~/lab1-repo]
$ curl -I http://192.168.56.103
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 18:52:18 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

Status code: 200 OK

Server header: Apache/2.2.8 (Ubuntu) DAV/2

Step 24: Basic Fingerprint

```
(kali@Fati)-[~/lab1-repo]
$ whatweb http://192.168.56.103
http://192.168.56.103 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.103], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

WhatWeb identified: Apache 2.2.8, running on Ubuntu Linux, PHP 5.2.4-2ubuntu5.10, WebDAV version 2, and the page title "Metasploitable2 - Linux". This matches everything found earlier with Nmap and curl, just presented in WhatWeb's own compact format.

Step 25:

```

(kali@Fati)-[~/lab1-repo]
$ whatweb -v http://192.168.56.103
WhatWeb report for http://192.168.56.103
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.103
Country     : RESERVED, ZZ

Summary      : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

Version      : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to identify the operating system from the server header.

OS           : Ubuntu Linux
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

Version      : 5.2.4-2ubuntu5.10
Google Dorks: (3)
Website      : http://www.php.net/

[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol (HTTP) that facilitates collaboration between users in editing and managing documents and files stored on World Wide Web servers. - More Info:

```

```

http://en.wikipedia.org/wiki/WebDAV

Version      : 2

[ X-Powered-By ]
X-Powered-By HTTP header

String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 18:58:20 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

```

The basic scan in Step 24 gave a short one-line summary with the main technologies found. This verbose version shows the exact same technologies (Apache, PHP, WebDAV, X-Powered-By), but adds a lot more detail — it explains what each plugin actually is, breaks down where the info

came from (like "from HTTP Server Header"), and shows the full raw HTTP headers at the end. Nothing new was found, just gave a lot more explanation and context around the same results.

Step 26: Aggression level 1

```
(kali@Fati)-[~/lab1-repo]
$ whatweb -a 1 http://192.168.56.103
http://192.168.56.103 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.103], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

Step 27: Aggression level 3

```
(kali@Fati)-[~/lab1-repo]
$ whatweb -a 3 -v http://192.168.56.103
WhatWeb report for http://192.168.56.103
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.103
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
    The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

    Version      : 2.2.8 (from HTTP Server Header)
    Google Dorks : (3)
    Website      : http://httpd.apache.org/

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to identify the operating system from the server header.

    OS           : Ubuntu Linux
    String        : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
    PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

    Version      : 5.2.4-2ubuntu5.10
    Version      : 5
    Google Dorks : (3)
    Website      : http://www.php.net/

[ WebDAV ]
    Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol (HTTP) that facilitates collaboration between users in editing and managing documents and files stored on World Wide Web servers. - More Info:
```

```
http://en.wikipedia.org/wiki/WebDAV
Version      : 2
[ X-Powered-By ]
X-Powered-By HTTP header
String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)
HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 19:24:43 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html
```

The only difference noticed was under PHP, where it now lists "Version: 5" as an extra line alongside the full version number. So going from level 1 to level 3 didn't really change much for this target, it just made a couple more requests and was more detailed.

Step 28: Aggression Level 4

```

(kali@Fati)-[~/lab1-repo]
$ whatweb -a 4 -v http://192.168.56.103
WhatWeb report for http://192.168.56.103
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.103
Country     : RESERVED, ZZ

Summary      : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], Matomo, PHP[5.5.24-2ubuntu5.10], WebDA
V[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
    The Apache HTTP Server Project is an effort to develop and
    maintain an open-source HTTP server for modern operating
    systems including UNIX and Windows NT. The goal of this
    project is to provide a secure, efficient and extensible
    server that provides HTTP services in sync with the current
    HTTP standards.

    Version      : 2.2.8 (from HTTP Server Header)
    Google Dorks: (3)
    Website      : http://httpd.apache.org/

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to
    identify the operating system from the server header.

    OS           : Ubuntu Linux
    String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ Matomo ]
    Matomo is the leading open alternative to Google Analytics
    that gives you full control over your data. Matomo lets you
    easily collect data from websites, apps & the IoT and
    visualise this data and extract insights. Privacy is
    built-in. Matomo was formerly known as Piwik, and is
    developed in PHP.

    Aggressive function available (check plugin file or details).
    Google Dorks: (1)
    Website      : https://matomo.org

[ PHP ]
    PHP is a widely-used general-purpose scripting language
    that is especially suited for Web development and can be
    embedded into HTML. This plugin identifies PHP errors,
    modules and versions and extracts the local file path and
    username if present.

```

```

    Version      : 5.2.4-2ubuntu5.10
    Version      : 5
    Google Dorks: (3)
    Website      : http://www.php.net/

[ WebDAV ]
    Web-based Distributed Authoring and Versioning (WebDAV) is
    a set of methods based on the Hypertext Transfer Protocol
    (HTTP) that facilitates collaboration between users in
    editing and managing documents and files stored on World
    Wide Web servers. - More Info:
    http://en.wikipedia.org/wiki/WebDAV

    Version      : 2

[ X-Powered-By ]
    X-Powered-By HTTP header

    String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 19:28:27 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

```

it detected Matomo that runs on PHP.

Step 29: Follow Redirects

```
(kali㉿Fati)-[~/lab1-repo]
$ whatweb --follow-redirect=always http://192.168.56.103
http://192.168.56.103 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.103], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

No redirect happened

Step 30:

```
(kali㉿Fati)-[~/lab1-repo]
$ whatweb -v http://192.168.56.103 > whatweb-results.txt

(kali㉿Fati)-[~/lab1-repo]
$ cat whatweb-results.txt
WhatWeb report for http://192.168.56.103
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.103
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
  The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

  Version      : 2.2.8 (from HTTP Server Header)
  Google Dorks: (3)
  Website      : http://httpd.apache.org/

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to identify the operating system from the server header.

  OS           : Ubuntu Linux
  String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
  PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

  Version      : 5.2.4-2ubuntu5.10
  Google Dorks: (3)
  Website      : http://www.php.net/

[ WebDAV ]
  Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol
```

```

(HTTP) that facilitates collaboration between users in
editing and managing documents and files stored on World
Wide Web servers. - More Info:
http://en.wikipedia.org/wiki/WebDAV

Version      : 2

[ X-Powered-By ]
X-Powered-By HTTP header

String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 19:41:55 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

```

Port	Protocol	State	Service	Version/Evidence	Command
21	TCP	Open	ftp	vsftpd 2.3.4	nmap -p- -sV
22	TCP	Open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1	nmap -p- -sV
23	TCP	Open	telnet	Linux telnetd	nmap -p- -sV
25	TCP	Open	smtp	Postfix smtpd	nmap -p- -sV
53	TCP	Open	Domain (DNS)	ISC BIND 9.4.2	nmap -p- -sV
80	TCP	Open	http	Apache httpd 2.2.8 (Ubuntu) DAV/2	nmap -p- -sV
111	TCP	Open	rpcbind	RPC	nmap -p- -sV
139	TCP	Open	netbios-ssn	Samba smbd 3.X-4.X	nmap -p- -sV
445	TCP	Open	netbios-ssn	Samba smbd 3.X-4.X	nmap -p- -sV

See step 11 for full list

Part 8

1. What is reconnaissance?

Reconnaissance is the first stage of a security test, where you gather information about a

target before doing anything else — like what's running, what version, what OS — without trying to break in yet.

2. Difference between host discovery and port scanning?

Host discovery just checks if a device is alive on the network at all. Port scanning goes further and checks which specific doors (ports) are open on one device you've already found.

3. Explain open, closed and filtered.

Open means something is listening and responded. Closed means the device replied but said nothing's using that port. Filtered means no reply came back at all, usually because a firewall is blocking it, so you can't tell if it's really open or not.

4. What does -sV do?

It tries to figure out exactly what software and version is running behind each open port, instead of just the generic service name.

5. What does -O do?

It tries to guess the target's operating system by looking at how its network responses behave.

6. What does -A combine, and why is it noisier?

It combines OS detection, version detection, default scripts, and traceroute all in one scan. It's noisier because it sends a lot more probes/requests than a plain scan.

7. What does -p- mean?

It tells Nmap to scan all 65,535 TCP ports instead of just the common ~1000.

8. Why can UDP scans take longer?

Because UDP doesn't have a handshake like TCP, so when there's no reply, Nmap has to wait out a timeout before moving on, and this happens a lot with UDP.

9. What does -sC do?

It runs Nmap's default set of NSE scripts, which dig a bit deeper into each service than a plain scan would.

10. What did the HTTP title/headers scripts reveal?

The title showed "Metasploitable2 - Linux," and the headers showed the exact web server (Apache 2.2.8) and that PHP was running behind it.

11. What did SMB/SSH/FTP enumeration add?

SMB showed the protocol version in use and that message signing was off. SSH showed the exact host key algorithms. FTP confirmed anonymous login was allowed.

12. Difference between Nmap service detection and WhatWeb fingerprinting?

Nmap service detection works across many different types of services (FTP, SSH, databases, etc.), while WhatWeb only looks at web servers specifically, but goes into more depth on things like the web tech stack, frameworks, and page content.

13. How did WhatWeb levels 1, 3, and 4 differ?

Levels 1 and 3 gave basically the same result. Level 4 was the only one that found something extra — it detected Matomo, an analytics tool, that the lower levels missed.

14. Why must aggressive scanning stay inside the authorised lab?

Because it sends a lot of probes that could look like an attack, and doing this against a system you don't have permission for would be illegal and could cause problems for that system.